

OpenDccCLib

Full NMRA DCC Protocol Stack in C

Runs on any processor. No OS. No heap. Just clean C. Edition 2026-09.

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What is DCC?

DCC (Digital Command Control) is the NMRA standard for controlling model railroad locomotives and accessories. A command station sends addressed speed, function and programming commands over the rails as a modulated square wave; decoders in locomotives and accessories receive and act on them. DCC covers short and long addresses, 128-step speed control, 68 function outputs, CV-based configuration, service-mode programming, and bidirectional communication through RailCom.

What is OpenDccCLib?

The full NMRA DCC protocol stack, written in plain C, for command stations and decoders.

"No dynamic memory. No OS. No external dependencies. If your chip has a C compiler and a timer, it can run OpenDccCLib."

Key Features

- **Zero heap.** Every buffer is a static array sized at compile time from `USER_DEFINED_*` constants.
- **Command station and decoder in one library.** Compile-time role flags select what you need; a booster can be both.
- **Full service mode.** Direct, paged, register and address programming, a mode-detect probe, ACK detection with configurable thresholds and a hardware-verified width window.
- **RailCom.** Command-station cutout timed by a five-state machine, armed at the end bit's last edge; 4/8 decode of Channel 1 and 2 replies, tagged with the packet's address. Decoder-side reply engine with bit-banged 4/8 transmit (the shipped decoder example leaves the transmit pin unwired).

- **Spec-correct repeats.** Every packet builder carries its repeat count, the standard's where it states one and a maintainer default where it is silent: two identical packets for a CV write, three for a date, one for a verify.
- **Dependency injection.** Hardware is reached only through function pointers wired at initialization. Port by rewriting the driver files for your timers, GPIO and UART; the protocol code is untouched.
- **Proven on the wire.** Beyond the unit tests, a Saleae hardware-in-the-loop bench checks the shipped firmware against the NMRA standards, down to microsecond timing and per-packet repeat counts.
- **BSD 2-Clause.** Use in open-source or commercial products, no royalties.

Protocol Coverage

Module	Details
Bit encoder	NMRA 58 μ s one-bit and 116 μ s zero-bit halves (one and two ticks) from a shared 58 μ s timer; single-buffered serialization from the ISR; RailCom cutout armed at the end bit's last edge
Packet builders	Speed 14/28/128, F0–F68, basic and extended accessories with CV access and NOP, consist, binary state, analog, time and date, loco CV access (POM); each with its spec-correct repeat count
Scheduler	Static slots; one-shots go first, ranked by priority (ESTOP > SPEED > FUNCTION > ACCESSORY > CV > IDLE); duplicate combining by (address, tag); two-tier paced auto-refresh (a prompt burst, then a keep-alive with a starvation bound); an idle spacer between back-to-back packets to the same short address 112–127
Service mode	Direct, paged, register, address primitives plus read/write/verify task orchestrators and mode detection; configurable ACK threshold, window and retries
RailCom cutout	DELAY 26, SETTLING 54, CH1 97, GAP 16, CH2 261 μ s (454 μ s total); each period settable per channel in <code>dcc_config_t</code> (0 = spec default) and adjustable at runtime; cancel and status calls
RailCom receive	Window-gated 250 kbaud UART hooks, standard S-9.3.2 4/8 code words, ACK and NACK, Channel 1 and 2 datagram assembly (split by byte count, so a Channel 2-only reply is currently read as Channel 1), tagging with the loco address of the preceding packet, receive ring
Bit decoder	Edge-timing classification with an 80 μ s threshold, stretched-zero tolerance, 10 ms no-signal recovery
Packet decoder	Preamble $\geq$ 10, byte assembly, XOR check, matching for short and long addresses (accessory commands are delivered for the application to match; consist addresses are not matched), deferred dispatch queue so command callbacks never run in an interrupt
CV storage	Function-pointer storage with decoder lock (CV 15/16), factory reset on CV 8, indexed CVs (CV 31/32, 257–512), CV 29 feature mask hook
RailCom transmit	Decoder reply engine: automatic Channel 1 address broadcast after this decoder's own commands, application-filled Channel 2 through <code>on_railcom_request</code> (called on the edge path), ACK/BUSY/NACK tokens; accessory SRQ and status helpers exist but are not yet wired into the transmit engine
Fail-safe	S-9.2.4 packet time-out from CV 11 in 100 ms units, with enter and exit callbacks

Example Platforms

Two ready-to-run projects ship with the library. Porting to a new platform means rewriting the driver files: timers and GPIO, UART, and for the decoder the ACK pulse.

Example	Platform	Toolchain
Command station	TI MSPM0G3507 LaunchPad	Code Composer Studio / Theia
Decoder	TI MSPM0G3507 LaunchPad	Code Composer Studio / Theia

Architecture Highlights

- One `dcc_config_t` struct, one `DccConfig_initialize()` call; the command-station API comes from three small `dcc_application_command_station_*.h` headers.
- Every module talks to its neighbours through an interface struct, so each is mocked in isolation and an MCU swap touches only the driver layer.
- A single `DccConfig_run()` per main-loop pass drives the whole engine; every application callback fires there, except the decoder's `on_railcom_request`, which answers on the edge path.
- A shared 58 μ s timer clocks both the main and the service track; the decoder's edge path classifies bits and assembles packets; the shipped example timestamps in the interrupt and drains in the main loop.
- Compile-time role selection: command station, decoder, accessory decoder, or any combination.

Test Coverage

Metric	Value, measured at generation (2026-09-25)
Unit-test binaries	29 GoogleTest binaries, one per module
Test cases	1246, all passing, zero warnings
Coverage (gcovr)	99.7 % lines, 100 % functions, 98.2 % branches
Compile-gate matrix	Six single-role configurations built and linked on every run
Source	60 files, 18 399 lines in <code>src/dcc/</code>
Hardware-in-the-loop	6 Saleae suites plus a 14-check bench preflight, 396 on-wire checks: S-9.1, S-9.2, S-9.2.1, S-9.2.3, S-9.3.2, scheduler (incl. refresh pacing); one expected FAIL records the Channel 2-only limitation

Getting Started

1. **Command station.** Import `applications/ti_theia/mspm03507_launchpad/command_station/` into CCS, build, flash, open a 230400-baud terminal and type `POWER ON`. DCC on the track in minutes.
2. **Decoder.** Import `.../decoder/`, build, flash, connect a DCC source. Every decoded command prints as a `RECV` line.
3. **New platform.** Copy an example, rewrite the driver files for your timers and GPIO. The protocol engine runs unchanged.

Documentation

- Quick Start Guide, Command Station and Decoder: step-by-step LaunchPad walkthroughs.

- Developer Guide, Command Station and Decoder: config struct, internals, porting, testing.
- `documentation/ARCHITECTURE.md`: the as-built design. `documentation/compliance/`: per-feature NMRA compliance status with machine-checked test references.
- API reference: Doxygen from the headers.
- github.com/JimKueneman/OpenDccCLib, BSD 2-Clause.

Built for model railroaders, by a model railroader.